

Department of Mathematics and Applied Mathematics
Module 3MS: Metric Spaces

Class Test One

29 March 2018

Time : 17h00 – 18h30

Total marks: 40

Please answer Questions 1 to 3 in one answer book, and Questions 4 to 7 in another.

Remember that when you give a counter-example, you must specify the metric as well as the set you are considering.

1. Let d_E be the Euclidean metric on $\mathbb{R}$ and d_D the discrete metric on $\mathbb{R}$.
Let $A = [0, 4) \cup (10, 20]$.

- (a) In $(\mathbb{R}, d_E)$ write down $\text{int}(A)$. (No proof required.)
- (b) In $(\mathbb{R}, d_E)$ write down $\text{bd}(A)$. (No proof required.)
- (c) In $(\mathbb{R}, d_D)$ write down $\text{int}(A)$. (No proof required.)
- (d) In $(\mathbb{R}, d_D)$ write down $\text{bd}(A)$. (No proof required.) [4]

2. Let $g : [0, \infty) \rightarrow [0, \infty)$ be a function such that these properties hold:
For all $a, b \in [0, \infty)$,

- (a) $g(a) = 0 \iff a = 0$
- (b) $a \leq b \implies g(a) \leq g(b)$
- (c) $g(a + b) \leq g(a) + g(b)$

Suppose (X, d) is a metric space. Define ρ by:

$$\rho(x, y) = g(d(x, y)) \text{ for all } x, y \in X$$

Prove carefully that ρ is a metric on X . Indicate clearly where each of the properties (a), (b) and (c) is used. [6]

3. For $(a, b) \in \mathbb{R}^2$ and $(c, d) \in \mathbb{R}^2$, define:

$$\delta((a, b), (c, d)) = \max \{|a - c|, 5|b - d|\}$$

You are given, and do not need to check, that δ is a metric on $\mathbb{R}^2$.

In the metric space $(\mathbb{R}^2, \delta)$, provide a sketch of $B((0, 0); 10)$, that is, the open ball centred at $(0, 0)$ with radius 10. Provide brief working to justify your sketch. [3]

Please change to a new answer book now.

4. Read the following theorem and proof, and then answer the questions that follow.

Theorem Let A be a subset of a metric space (X, d) .

Suppose $x \in A$ and, for all $r > 0$, $B(x; r) \cap A' \neq \emptyset$.

Then $x \in \text{bd}(A)$.

Proof Suppose $x \in A$ and, for all $r > 0$, $B(x; r) \cap A' \neq \emptyset$. (1)

Then, for all $n \in \mathbb{N}$, $B(x; \frac{1}{n}) \cap A' \neq \emptyset$. (2)

Then, for all $n \in \mathbb{N}$, there exists $z_n \in B(x; \frac{1}{n}) \cap A'$. (3)

Then $z_n \rightarrow x$ as $n \rightarrow \infty$. (4)

It follows that $x \in \overline{A'}$. (5)

So $x \in \text{bd}(A)$. (6)

- (a) Write down the definition of the boundary of a subset A of a metric space.
- (b) Explain how line (4) follows from the previous working.
- (c) Explain how line (5) follows from the previous working.
- (d) Explain how line (6) follows from the previous working. [8]

5. (a) Prove or disprove: If F and K are closed subsets of a metric space, then $F \cap K$ is a closed subset of the metric space.

- (b) Prove or disprove: For any subsets A and B of a metric space, $\overline{A \cap B} = \overline{A} \cap \overline{B}$. [7]

6. Let (X, d) be a metric space, $x \in X$ and A a non-empty subset of X .

- (a) Write down the definition of $\text{dist}(x, A)$. (Also sometimes denoted $d(x, A)$.)
- (b) Write down the definition of the closure of a subset, in the form:
“ $x \in \overline{A} \iff \dots$ ”
- (c) Prove that if $\text{dist}(x, A) = 0$ then $x \in \overline{A}$, using the definitions above (and no other method). [8]

7. (a) Suppose that (X, d) is a metric space in which every sequence has a convergent subsequence. Prove that there exists $x \in X$ and $r > 0$ such that $B(x; r) = X$.

- (b) Is it possible to find a metric space (X, d) such that every subset of X is open or closed, but not all subsets of X are clopen (i.e. closed and open)?

If so, provide such a space, justifying your answer; if not, explain why not.

[4]